

	and hence the <u>answer in (iii) would be less.</u>	[1]
7(a)	Electric field strength $E = \frac{V}{d} = \frac{40 - (-40)}{40 \times 10^{-3}} = 2.0 \times 10^3 \text{ V m}^{-1}$	[1] answer
7(b)	Acceleration $a = \frac{F}{m} = \frac{eE}{m} = \frac{(1.6 \times 10^{-19})(2.0 \times 10^3)}{9.11 \times 10^{-31}}$ $= 3.51 \times 10^{14} \text{ m s}^{-2}$	[1] substitution [1] answer (2 or 3 s.f.)
7(c)	Duration of time for electron to stay in electric field is given by $t = \frac{s}{v} = \frac{80 \times 10^{-3}}{1.5 \times 10^7} = 5.333 \times 10^{-9} \text{ s}$	[1] for value of t

	Vertically, using $v = u + at$, $v = 0 + (3.51 \times 10^{14})(5.333 \times 10^{-9})$ $= 1.87 \times 10^6 \text{ m s}^{-1}$ $= 1.9 \times 10^6 \text{ m s}^{-1}$	[1] for velocity $1.87 \times 10^6 \text{ m s}^{-1}$
7(d)	$\tan \theta = \frac{v_y}{v_x} = \frac{1.9 \times 10^6}{1.5 \times 10^7} \rightarrow \theta = 7.2^\circ$	[1] answer
7(e)(i)	Magnetic force provides centripetal force $Bev = \frac{mv^2}{R}$ $\rightarrow R = \frac{mv}{Be} = \frac{(9.11 \times 10^{-31})(1.9 \times 10^6)}{(1.62 \times 10^{-4})(1.60 \times 10^{-19})}$ $= 6.68 \times 10^{-2} \text{ m} = 6.7 \times 10^{-2} \text{ m}$	[1] statement [1] substitution [1] answer
7(e)(ii)	$Bev = \frac{mv^2}{R} \rightarrow Be = \frac{mv}{R} = m\omega = \frac{2\pi m}{T}$ $\rightarrow T = \frac{2\pi m}{Be} = \frac{2\pi(9.11 \times 10^{-31})}{(1.62 \times 10^{-4})(1.60 \times 10^{-19})}$ $= 2.21 \times 10^{-7} \text{ s} = 2.2 \times 10^{-7} \text{ s}$	[1] substitution [1] answer
8(a)(i)	$20.0 \text{ km s}^{-1} = 2.0 \times 10^4 \text{ m s}^{-1}$	